

# Generation Interconnection System Impact Study Report

## For Queue Project Q20-1391

Thompson Peak Tap 138 kV

January 17, 2022

### 1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q20-1391. The Interconnection Customer proposes a 145.1 MW solar photovoltaic facility interconnecting at the Thompson Peak Tap 138 kV substation in the Maryland area. The request was received on January 19, 2020. The study evaluated system intact and single contingency conditions using the regional planning case, including 356 higher-queued active requests and their assigned network upgrades.

### 2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 10 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$7,751,000.

### 3. Network Impacts

#### 3.1 Generator Deliverability and Single Contingency Analysis

| Monitored Facility                                     | Contingency                                 | Loading (%) | Rating (MVA) | DFAX (%) |
|--------------------------------------------------------|---------------------------------------------|-------------|--------------|----------|
| Thompson Peak Tap – East Towson 138.0 kV               | Coldspring – Recovery Soco 138.0 kV         | 148.2       | 279          | 21.7     |
| Center – E Towson 138.0 kV                             | Coldspring – Recovery Soco 138.0 kV line    | 135.4       | 400          | 21.0     |
| Coldspring – Ctr Substation 138.0 kV                   | East Towson – Ctr 138.0 kV                  | 130.5       | 347          | 12.1     |
| Recovery Soco – Coldspring 138.0 kV L                  | Ctr – East Towson 138.0 kV Line             | 128.6       | 403          | 12.1     |
| E Towson – Lone Peak 138.0 kV line                     | Thompson Peak Tap – E Towson 138.0 kV       | 126.0       | 193          | 14.8     |
| Center – Highgrove 138.0 kV Line                       | Sierra Pacific Lumber – Riva Road 345.0 kV  | 117.3       | 519          | 18.0     |
| Ramos Substation – Danskammer Generating Sta 138.0 kV  | Highgrove – Ctr 138.0 kV                    | 112.8       | 265          | 9.1      |
| Westport – Warren 138.0 kV Line                        | Sierra Pacific Lumber 138 kV/1 kV TX        | 110.6       | 318          | 6.4      |
| Danskammer Generating Station – Painters Mill 138.0 kV | Ctr – Highgrove 138.0 kV Line               | 109.0       | 250          | 9.1      |
| Dpc New Lisbon – Thompson Peak Tap 138.0 kV Line       | Hidalgo Smelter 345 kV/1 kV Autotransformer | 105.3       | 275          | 12.8     |

#### 3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

#### 3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

#### 4. Required Network Upgrades and Cost Estimates

| Facility                                               | Required Upgrade                            | Cost Estimate | Allocation to Project |
|--------------------------------------------------------|---------------------------------------------|---------------|-----------------------|
| Thompson Peak Tap – E Towson 138.0 kV                  | Rebuild line with higher-capacity conductor | \$2,637,000   | 42% (\$1,121,000)     |
| Center – E Towson 138.0 kV                             | Rebuild line with higher-capacity conductor | \$9,050,000   | 18% (\$1,632,000)     |
| Coldspring – Ctr Substation 138.0 kV                   | Rebuild line with higher-capacity conductor | \$933,000     | 10% (\$92,000)        |
| Recovery Soco – Coldspring 138.0 kV                    | Rebuild line with higher-capacity conductor | \$2,751,000   | 10% (\$270,000)       |
| E Towson – Lone Peak 138.0 kV line                     | Rebuild line with higher-capacity conductor | \$5,343,000   | 39% (\$2,074,000)     |
| Center – Highgrove 138.0 kV Line                       | Rebuild line with higher-capacity conductor | \$2,901,000   | 13% (\$371,000)       |
| Ramos Substation – Danskammer Generating Sta 138.0 kV  | Rebuild line with higher-capacity conductor | \$3,250,000   | 36% (\$1,170,000)     |
| Westport – Warren 138.0 kV                             | Rebuild line with higher-capacity conductor | \$1,999,000   | 18% (\$370,000)       |
| Danskammer Generating Station – Painters Mill 138.0 kV | Reconductor line                            | \$820,000     | 36% (\$295,000)       |
| Dpc New Lisbon – Thompson Peak Tap 138.0 kV Line       | Reconductor line                            | \$1,078,000   | 33% (\$356,000)       |
|                                                        | Total allocated to project                  |               | \$7,751,000           |

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.